

A solution for Midband

A J Dickie and J H Mackenzie

Midband is the BT name coined for the platform which is intending to deliver new data services with access rates in the range 200 kbit/s to approximately 8 Mbit/s over existing copper infrastructure. The technology which is perceived to make this possible is ADSL (asymmetric digital subscriber line) which is one of a number of new and exciting advanced line coding techniques which offer the tantalising prospect of being able to provide broadband bandwidths over existing metallic access networks. ADSL is seen as offering cost-effective high-speed network access where none has existed previously, and as such offers a significant opportunity for those operators with an installed copper access network to further exploit that asset. Other alternatives for providing high-speed access are less likely to be successful as they rely on replacing a significant portion of the current copper access network with fibre which is currently seen as prohibitively expensive. Therefore, the view often offered from many analysts is that this technology will be made to work because it offers the only viable way forward. Certainly the amount of effort, the trials, the standards and the new variants on this technology suggest this to be the case, but this also suggests that the technology is still maturing and that it would be unwise at this stage to bet on the final solution. There has already been much written on this subject and it would be easy to be swayed into believing that all the problems have been solved and that it is simply a matter of now rolling out this technology. This paper attempts to show that, while ADSL access is the most promising proposition, there is still a significant amount of work to be done in order to make the case for mass deployment within the UK.

1. What is Midband?

Midband is the name of one of three platforms created by BT to deliver higher capacity digital services. Each platform is differentiated by the access technology used and the associated network capability. The three platforms are:

- Lowband (HomeHighway), based on ISDN and delivering symmetrical 2B plus 1 D channel to the end user — it uses the PSTN network for the core,
- Midband, based on ADSL [1] and delivering up to 8 Mbit/s to the customer with around 10% bandwidth in the return direction — over longer distances the capability reduces to several hundred kbit/s downstream,
- Highband, based on VDSL (very high bit rate digital subscriber line) [1, 2] and able to offer symmetrical or asymmetrical services — in asymmetrical mode it can deliver up to 25 Mbit/s downstream over distances up to 1 km.

Midband makes use of the capabilities of ADSL to attack the broadband market using exchange-based ADSL equipment. In this way, capabilities of several Mbit/s are possible from existing buildings with minimal access network infrastructure upgrade.

All of the above access systems are based on available or soon to be available technology and comprehensive

technical descriptions of this technology are detailed elsewhere in this Journal. In simple terms it is accomplished by providing a digital modem at each end of the copper line to provide the high bit rate channel. Figure 1 gives an indication of the capabilities of some xDSL technologies, and as can be seen performance is highly length dependent. In addition, the performance is highly dependent upon the quality and gauge of the metallic network as well as interference from other transmission systems on neighbouring metallic pairs sharing the same cable. ADSL and VDSL technologies also allow existing POTS services to be maintained on the same metallic pair because these technologies use unallocated frequency spectrum in the copper transmission line. To achieve this an active spectral splitter is required at the exchange and customer premises end to separate the POTS and xDSL signals.

2. Target market and services

Within BT, Midband was originally conceived for use as a defence to the competitive threat of cable modems. The major factors were that in order to compete it should be able to offer at least as good a service, at a competitive price. Looking wider afield, xDSL technology offers the potential to attack many market sectors and deliver multiple types of services each with different transport requirements.

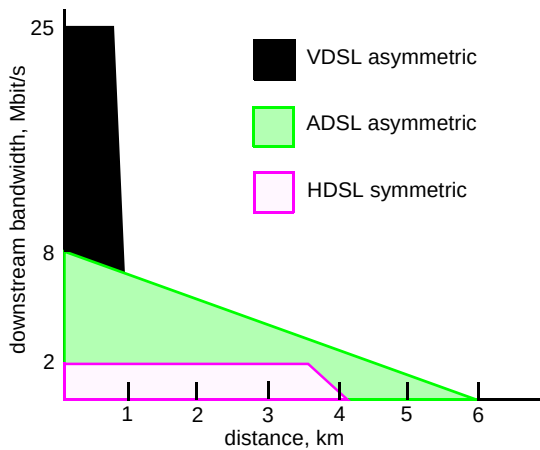


Fig 1 Outline xDSL capability

Many published reports and studies exist and a typical survey is shown in Fig 2 [3]. The types of services and applications which are being considered are:

- residential
 - entertainment,
 - home shopping/banking,
 - games,
 - high-speed Internet,
- business
 - small office home office (SOHO),
 - small-to-medium enterprises (SME),
 - teleworking/LAN interconnect,
 - videoconferencing,
 - support legacy services (FR, SMDS, etc).

Each of these applications will place different requirements on the transport network in terms of bit rate, delay, reliability, burstiness, interactivity, etc. What is clear is that the network should be capable of supporting a wide

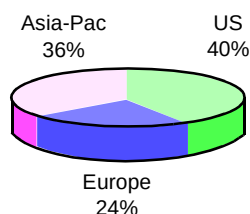
variety of services with different quality of service guarantees, and it should also be capable of supporting voice, video and data transport (i.e. support the move towards multimedia). It is also apparent that there does not appear to be a single market sector or application that is seen as the 'killer application'. However, many observers are convinced that many of the applications will be based on the IP protocol and thus the carriage of IP on ADSL-based networks is essential.

BT sees the near-term use of this technology not only for providing high-speed Internet access but also for connecting remote offices and providing corporate IP networks. The ADSL technology is configured to allocate higher bandwidths to the downstream direction as it is expected that the most obvious use for this technology is for connection to the Internet where most users will be downloading much more information than uploading. IP access today provides no QoS guarantees. Being packet based, the network usage can be very bursty with sustained throughputs of a few kbit/s. Currently, IP networks are designed for contention ratios around 50:1. New services will mean that this will change as the applications are becoming more bandwidth hungry and require improved performance (delay, throughput, packet loss, etc). Many ADSL vendors are standardising on the use of ATM-based systems which allows the equipment to support services which need differentiated QoS levels.

3. What's in it for the telco?

The cost of upgrading the access network has hitherto been seen as a major barrier to the economic deployment of broadband services. Figure 3 illustrates the current BT narrowband network based on transit, local and remote concentrating PSTN switches. In all, BT has around 5500 operational buildings across the UK supporting 28 million working PSTN lines. On average a customer is less than 3 km from the nearest BT building although this can vary

world xDSL market in 2003
lines = 19 m
revenue = \$16.5 bn



xDSL access lines by customer (000s)
Europe

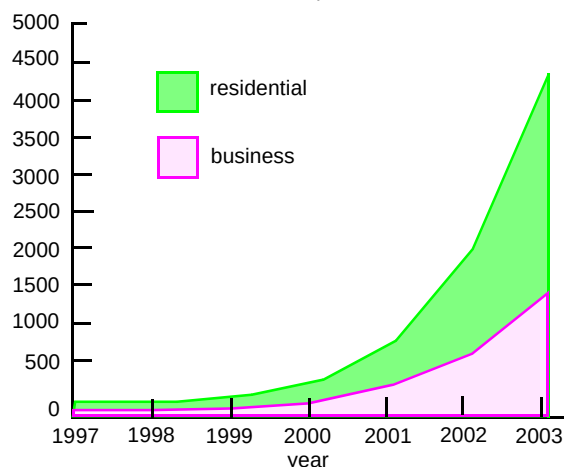


Fig 2 Potential world market for xDSL.

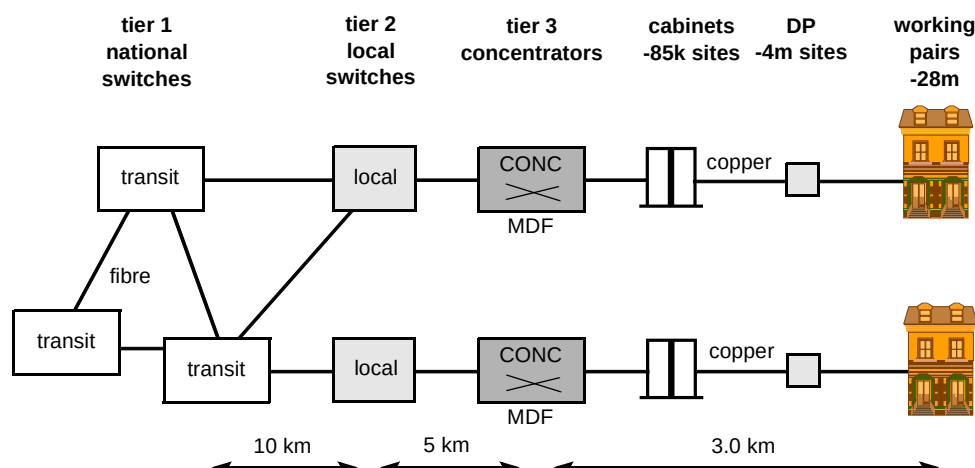


Fig 3 Existing BT narrowband network.

significantly. This network has evolved over several decades and takes huge capital investment. BT spends around £3b [4] capital expenditure every year in upgrading and extending this network.

For the average data user the ubiquitous PSTN modem operating at speeds of tens of kilobits has been the only economic option widely available. It is apparent that, as the bandwidth requirements escalate due to new applications, PSTN modems are incapable of meeting these demands. Traditionally copper has not been capable of delivering the high access speeds and so alternative access technologies such as direct fibre, PONs and fibre to the exchange/cabinet have been considered but this requires new infrastructure build. Clearly any new access technology which requires any form of infrastructure upgrade would take many years to roll out widely and require significant capital investment, which, in today's competitive market place, telcos cannot support, particularly when the services are still immature. Consequently xDSLs which reuse the existing infrastructure relieves one of the major barriers to the widespread deployment of new services for incumbent telcos with an installed metallic network. The potential advantages can be summarised as follows:

- reuse of existing infrastructure — takes a large capital asset which is used for a few minutes a day and enables additional revenue earning services to be delivered,
- speeds up the deployment of new services across the network as massive infrastructure build is not required,
- existing customer base provides a ready target market for new services.

From the viewpoint of BT this is a tremendous opportunity to exploit its copper access assets and to provide high-speed services to its customers much faster than had previously been thought possible. However, it should be stressed that, although there is significant potential with these technologies, there are still many

problems to be resolved, many of which are highlighted in this journal. Customer expectation must be carefully managed as the copper network is a variable entity and the characteristics of two neighbours' lines may be very different and thus support different access bandwidths. Although the technology is capable of delivering up to around 8 Mbit/s this is highly dependent on the copper network and the type of service being offered.

4. Simple end-to-end architecture

It should be recognised that although the access network forms the largest part of the cost burden of services (approximately two-thirds of the cost), the whole end-to-end architecture needs to be considered to engineer the appropriate network to meet the service requirements. One such architecture using ADSL is shown in Fig 4.

In general, the end-user's CPE could be a PC or set-top box if entertainment services are being provided. A number of different user network interfaces could be used including ATM25 and Ethernet. There is then the ADSL modem or ATU-R which superimposes the ADSL signals on to the copper pair. There is also a POTS splitter which may be physically separate from the ADSL modem to provide isolation between the POTS signals and the xDSL signals. The DSL access mux (DSLAM) multiplexes a number of DSL signals together on to a single physical interface which is presented to the core network. The core network generally consists of a number of ATM switches interconnected by SDH transmission equipment to the service providers. The connections across the ATM network tend to be permanently set up, i.e. permanent virtual circuits on a per-customer basis. The service provider (SP) connection could be any high-speed access rate — E1, E3 or STM-1.

For some time there has been a heated debate on the protocol stacks used on ADSL access; it has mainly centred

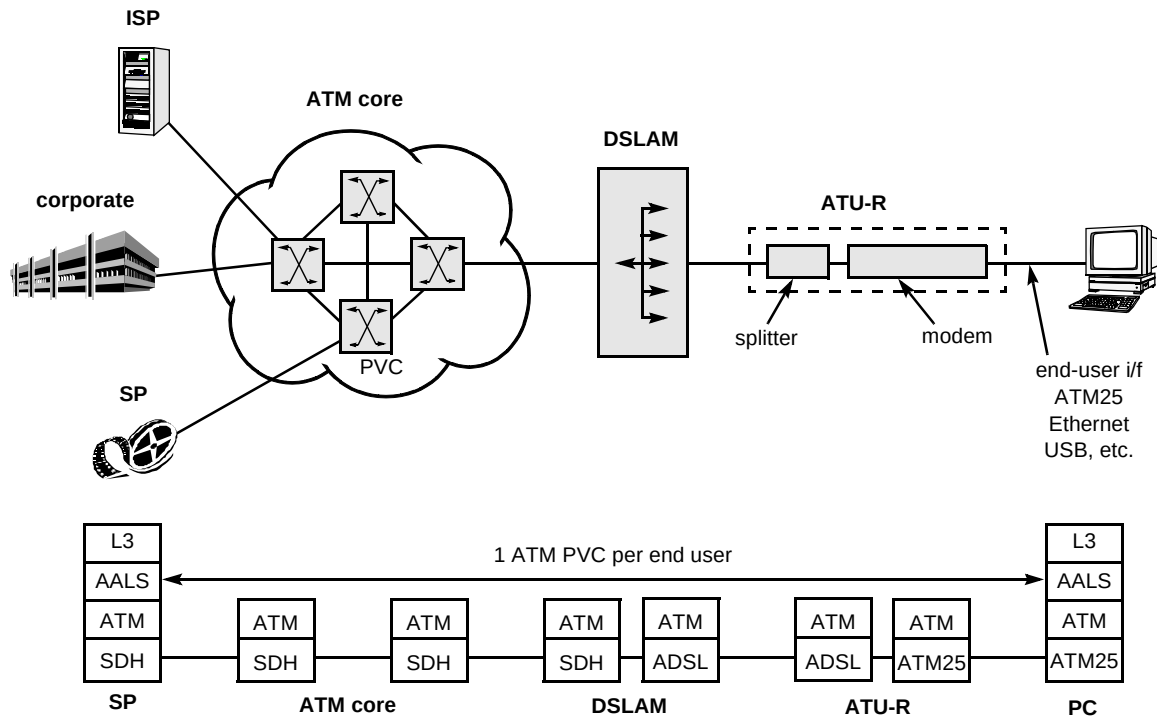


Fig 4 Typical end-to-end architecture for ADSL access (for simplification, only the protocol stacks associated with the ATM25 end-user interface are shown).

on whether the DSL bit pipe should be based on carrying ATM cells and then all services including IP are mapped into the ATM cells. Alternatively, IP packets are transported directly across the DSL bit pipe. On the one hand, as many of the applications are IP-based, it may initially be more bandwidth efficient to map IP packets directly on to the DSL. However, ATM offers the ability to guarantee different levels of quality of service with low latency. There is no clear-cut advantage one way or the other as developments in both technologies are moving to overcome their current drawbacks; however, what can be seen from the standards forums and products that are emerging is that the favoured architecture appears to be one based on carrying ATM cells as shown in Fig 4. The apparent drivers for this are that ATM is seen to offer greater transparency to different service requirements and that many telcos are already deploying ATM core networks for premium business services. Taking ATM all the way to the customer's premises enables the systems and processes developed for the emerging broadband core networks to be reused.

However, new architectures are being considered which are not solely based on ATM and may include some layer 3 functionality such as IP [5]. The proposals are aimed at making better use of core network bandwidth by using contention in layer 3 (as opposed to layer 2 in an ATM-based transport network) and moving towards a connection model similar to the dial-IP networks. However, this will involve the network provider in layer 3 systems management and processes, such as authentication, billing,

address management and session control, which will add additional complexity.

As well as looking at the physical boxes and how they are connected together, Fig 4 illustrates the protocol stacks and therefore the processing which needs to be performed within each equipment.

For simplification, only the protocol stacks associated with the ATM25 end-user interface are shown, while the interconnection with the narrowband network which is also using the same metallic pair is not shown.

5. It's not just about technology

In order to assess the viability of launching a service it is necessary to examine many more issues than just the technology even though that is vital to the overall equation. The most important aspect is ensuring that the business case is viable and this requires the comparison of whole life costs against the potential revenue streams. A firm understanding of the costs is crucial as this will determine how much the network will cost to build and run, which in turn will set the tariffs and have an impact on the target market and volumes. This is a very complex and highly linked problem space. If the costs are too high, tariffs have to be set at a level which may be too high to capture the target market. This in turn means that all of the target market may not be captured causing a drop off in revenue which then leads to a failing business case. Alternatively if the costs have been analysed thoroughly and optimised wherever possible then a cost

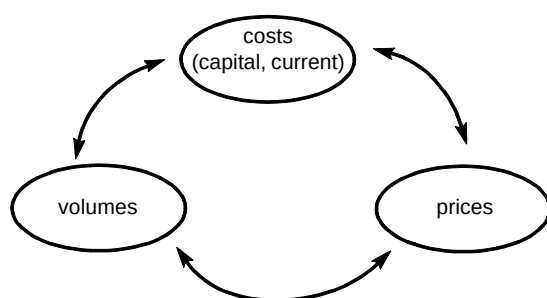


Fig 5 The virtuous or vicious business case circle.

base can be set which is acceptable to the target market and so leads to a positive business case (Fig 5).

Therefore the cost base must be examined in great detail and this includes many more aspects than the raw equipment costs. The whole-life costs [6] include:

- capital — up-front investment in network infrastructure and equipment and systems normally in advance of connecting a customer,
- non-recurring current account — the one-off cost for connecting customers as they take up the service (e.g. line cards, NTEs, accommodation refurbishment, training, customer installation, etc),
- recurring current account — the cost associated with running the network (plan, maintain, restore, churn) applies to every customer and it is an ongoing cost.

5.1 Minimising impact on existing services

The inherent assumption behind DSL technologies is the reuse of existing metallic infrastructure in order to reduce costs. However, this does bring with it an associated risk of affecting the quality and performance of the existing mature services and needs to be actively minimised. BT has installed and operates a comprehensive set of support systems and processes for existing services. These may need to be modified to accommodate the new ADSL technology and this can be a major problem if the new technology has been developed without taking the impact on existing services into consideration. This can add to the complexity and functionality of the ADSL equipment and needs to be factored in to the development programme early in the life cycle. For example, the purpose of the POTS splitter is to provide isolation between the POTS and ADSL signals so that the telephones do not get unacceptable noise as a result of the ADSL signal and that the ADSL modem is not corrupted by transient on/off hook impedance spikes. It is also necessary to ensure that existing processes for existing services are still possible, for example comprehensive automated fault diagnostics tools are available for POTS, such as remote line test. ADSL modems can interfere with existing POTS line test equipment and so it is necessary to provide some way of

remotely isolating the ADSL modems from the copper line while the POTS line test is being carried out.

5.2 Volume unit costs

Purchasing global standard products from a world market is essential to driving down unit costs. Development costs amortised over world market volumes rather than a specific telco's equipment requirements is crucial to reducing unit costs. However, all manufacturers offer slightly different products and the trick is to ensure that what is purchased can meet requirements with the minimum of further development. Telcos must exercise extreme care when asking for new and enhanced functionality as it may be deemed 'company special' where the telco may have to pay for the full development cost.

Working closely with vendors of such equipment is one way of ensuring that individual needs are best met. However, there are always ongoing developments and changes in market predictions which can easily upset this balance, and it is becoming increasingly necessary to ensure that vendors are able to bend with these requirements as necessary. In addition the BT access network is in many respects different from others in Europe and America which means the goal of 'global product' can easily be compromised.

It is also essential to understand when market take-up occurs and where potential technology breakpoints (and hence cost reduction) occur. The volume/unit price curve is closely linked to this and plays a significant part in the business case and any trade-offs to be made with the operational side of the equation. If this is gauged incorrectly then this will have a major impact on profit margin and the viability of the service.

5.3 Up-front development investment versus manual processes

There is often trade-off to be made between the cost and time to develop additional features and the consequential higher operational costs if these automated features are not developed. For example, development of automated configuration and management systems can greatly improve the speed of response of processes and reduce manual intervention. This is a direct balance between up-front investment and ongoing recurring costs — the decision is largely driven by volumes/time to market. It is important therefore that this is well understood early in the launch life cycle in order to ensure the necessary developments are put in place. If this does not happen soon enough, the programme inevitably has to compromise, which generally results in lower efficiency, increased cost and delay.

5.4 Competitive supply and convergence

If more than one ADSL equipment supplier is being used, there will be consequential costs if their offerings are not fully compatible. In general, it is better for BT to have more than one supplier in order to promote competition. However, in order to fully realise the benefit the supplier's equipment must have similar functionality and have some degree of interoperability, otherwise processes and network capability would need to support different solutions or lowest common capability. Significant time, effort and money must be invested by all parties (vendors and operators) to ensure that equipment offerings are functionally and operationally alike and deliver the same performance. This is particularly true of line coding mechanisms and element manager solutions.

It is unlikely at this stage that equipment supplied by any two vendors will interoperate, e.g. be able to share line cards, NTEs or use the same element manager. Even if there was a convergence plan this is unlikely to be achieved in short time-scales and is unlikely to reach full interoperability because of commercial drivers facing the equipment vendors. The current ADSL standards and variation in the way functionality in the DSLAM is provided by different suppliers ensures that full convergence is not likely, in which case there will be the added cost of operating two solutions in the network. This will incur extra training and carrying of more spares than necessary.

5.5 Technology-specific processes

With all new technologies there are a number of processes which are specific to that technology as a result of inherent features and this is no different when using ADSL. These can incur significant manual intervention and costs. The following are some of the major considerations when installing ADSL.

Customer identification

ADSL technology cannot reach all customers with the same bit rate and in some cases it is not even capable of delivering a minimum service to some customers. A key issue when a customer requests service is that a process needs to be started to evaluate the ability to deliver service to the customer. This is a very complex issue. It requires knowledge of the specific infrastructure to the customer and, for most telcos, this is stored in paper records which require manual assessment. Even then there will be a significant proportion of customers where a clear cut 'yes/no' answer is not possible. In this case, additional checks including possibly going into the network to physically test the performance could be required and this can be expensive. It is complicated by the fact that ADSL can be operated in two modes as shown in Fig 6.

- The rate adaptive mode is where the ADSL modem operates near its optimum limit to maximise the bandwidth delivered to the end user. In this way it should be possible to provide a degree of service to the vast majority of customers. However, this leads to customers having different bandwidths which will change with time as the amount of interference in the same cable changes. In general, the offered bandwidth will reduce with time. It is also difficult to determine the performance which the customer will receive and this can lead to problems with dimensioning the core network for a guaranteed QoS level. There are also implications on billing as it is difficult to accurately tariff for a capability which changes with time and could be different for every end user.
- The preferred BT approach is to operate the ADSL at a fixed bit rate, in order that some degree of performance is guaranteed even though this means it is not possible to reach all customers. To alleviate this it is possible to set up a limited set of fixed bit rates to ensure that the majority of customers can be reached.

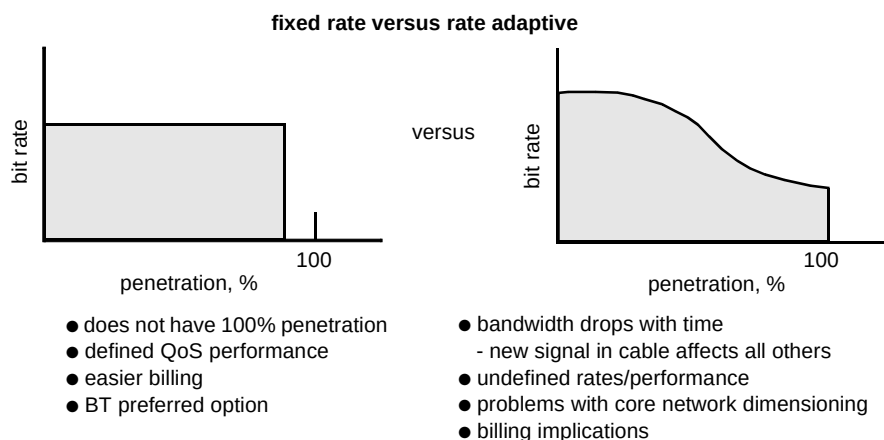


Fig 6 Fixed rate versus rate adaptive ADSL operation.

Incompatibility with other transmission systems

Although ADSL technology is compatible with existing POTS services there are a number of others with which it cannot share the same metallic pair. This is largely due to overlaps in the spectrum used for each of the different services. The major services are those based on ISDN and HDSL access, although there are others such as Telecom Red. Before ADSL access can be confirmed to the end user, it is necessary to establish the current services delivered over the customer's cable and to determine whether there is any incompatibility. This can seriously impact on costs and may even result in a second copper line being installed.

It is clear that a comprehensive access frequency plan is required where the spectrum used by each service is clearly understood. In this way future technologies can be developed where service incompatibility can be actively minimised.

Loop conditioning

In order for ADSL to work efficiently the spectrum which the system uses needs to be electrically 'free' from interference. Within the BT network there are a number of components currently used within the infrastructure which need replacing/upgrading to give acceptable DSL performance. Although the changes are relatively straightforward, it is not a simple process to identify those lines which have these components and it can take a considerable amount of time to identify the reason behind an ADSL line performing poorly.

6. What is on the customer's premises?

The interface presentation to the end user plays a significant role in the cost and marketing of the services. End users tend to want existing interfaces which can connect to their existing CPE while network operators may want new types of interface which simplify the network and move

functionality towards the service provider or end user. There are a number of interface options being considered (Fig 7) which can impact on the implementation of the network terminating equipment (NTE):

- ADSL modem with Ethernet interface

This presents an Ethernet 10BaseT interface to the end user which is a legacy interface for many Internet users today and therefore should be easier to market. Many users already self-install Ethernet network interface cards (NIC) costing a few tens of pounds in their own PCs. Today NTEs which present an Ethernet interface are unlikely to process IP packets and simply encapsulate the Ethernet frames into ATM cells (bridging) which requires the IP service provider to perform the unpacking in his equipment. This basically means that the end-users LAN is extended to the SP premises and can lead to configuration problems. The additional processing required in the SP equipment can limit scalability. A more elegant approach is for the NTE to process at the IP layer (layer 3) and become a simple router. Potentially, this can provide all the functions like network address translation, authentication and session control, but at additional cost and complexity. It is not yet clear what additional premium this could add to the cost of the NTE. Using an Ethernet presentation may present performance problems in that the Ethernet segment, being a shared medium, can give rise to variable delays and throughput. The ATM layer is terminated in the NTE and so it is not possible to provide ATM QoS guarantees.

- ADSL modem with ATM25 interface

This presents an ATM Forum 25 Mbit/s interface to the end user and is widely available from many DSL vendors today. As the DSL is carrying ATM cells, an ATM-based interface to the end user makes the NTE very simple. The modem receives cells from the PC

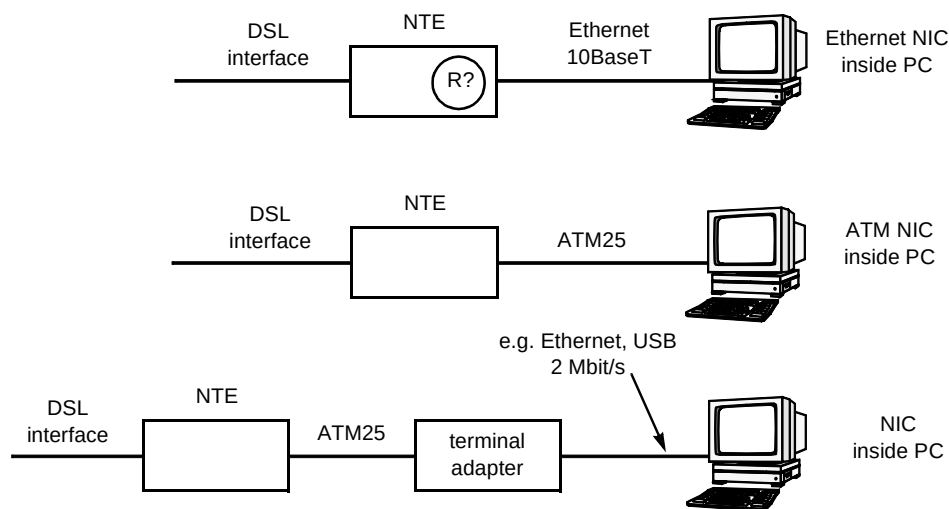


Fig 7 Options for the equipment on the customer's premises.

and transmits them across the ADSL link, effectively acting as a cell buffer to decouple the transmission rates between its two ports. It also extends the capabilities of ATM including its QoS guarantees all the way to the customer's CPE. As with the Ethernet interface it needs a NIC card to be installed in the customer's CPE. These cards are becoming widely available and their prices are dropping, but they are still more expensive and not so easy to install as Ethernet NIC cards. The ease of installation is very important as it is expected that the service provider would not want to get embroiled in setting up every customer's PC with the myriad of individual set-ups which are possible.

- ADSL modem with ATM25 interface plus external terminal adapter (TA)

A single end-user interface is unlikely to be accepted by all customers and so this approach provides flexibility. For those customers who need a different interface, perhaps because of existing legacy equipment, an external terminal adapter could be used to provide the interface adaptation. The functionality of the TA would depend upon the type of interface needed by the customer. Intuitively, several boxes on the customer's premises are more expensive than a single box but this would only apply to those customers who cannot accept an ATM25 interface.

- ADSL modem with universal serial bus (USB) interface

The universal serial bus is a standard interface which is widely available on computers today. It therefore does not require a NIC card with the associated cost and potential problems of configuration within the customer's PC. However, standards have yet to be defined on the methods for carrying IP and ATM over this interface.

As can be seen from the above the choice of what to offer is not straightforward and considerable thought must go into this decision. Even so, the market is very fickle and it is likely to change in short periods of time. This makes it critical that a flexible solution is provided which can cope with these unexpected changes.

From an access network perspective the drivers for the NTE and end-user interface are:

- the NTE provides the NTP which is the regulated boundary between the network and the end user,
- the NTE should be as simple as possible, to reduce cost and keep reliability high — this is particularly important for mass roll-out where the reliability of the NTE can have a major effect on maintenance costs,

- it should be service transparent to prevent built-in obsolescence — this is very important as it is not possible to change NTEs in short times-scales when new services are launched,
- the NTE must be easy for 'non-technical users' to install and manage — the aim is that the NTE can be installed/repared with little or no networking knowledge or background and that set-up requires little or no manual configuration (with zero manual configuration as the ultimate goal).

The pressure to keep the NTE cost low is a key driver, in particular with new ADSL-based broadband services as it is expected there will be a significant amount of churn of the customer base. This can be seen from many new services which have been launched recently, such as cable TV services where many operators experience 20% of their customer base changing service on a yearly basis.

When a customer ceases service, if the NTE is relatively inexpensive, it can be left *in situ*. However, if the NTE cost is high, a balance must be struck between writing-off a capital asset, and the cost of a customer visit to recover and refurbish the NTE so that it can be reused for another customer. Whatever the policy adopted, if there is a large degree of customer churn, then the cheaper the NTE and the easier it is to make a positive business case.

7. What next?

The capability of DSL equipment is evolving rapidly. Initially, the major effort was to verify that the basic technology actually worked and the many trials that have taken place throughout the world have proven this. However, there is still some way to go before equipment that is truly standards-compliant becomes available. Vendors are now beginning to concentrate on refining the DSL performance (reach/bit rate), increasing functionality and reducing costs. Other variants of DSL, such as DSL-Lite (or Splitterless DSL) are also being examined for potential cost reduction. Table 1 illustrates the potential developments within the DSLAM.

Table 1 DSLAM evolution.

	Today	Tomorrow
Network i/f	STM-1	STM-N, T/E3
Copper i/f	ADSL	Multiple DSLs
ATM QoS	Low	Multiple traffic class, shaping and policing
ATM multiplexing	Limited VP/VC	Full flexibility VP/VC
Connections	PVC	SVC (VB5.2?)
OAM	Limited	Comprehensive

Other key sectors of development include the following five areas.

- DSL-Lite or Splitterless DSL

This is based on standard DMT but does not include a splitter to provide isolation between the POTS and ADSL spectra. It is aimed at the high-speed Internet user or teleworker market and is being presented as analogous to the PSTN modem model where the end user could purchase the modem from a third party and plug it into the existing telephone socket. The cost of a 'truck roll' by the telco to the customer's premises is thereby avoided as is the cost of the splitter components. DSL-Lite will have significantly lower bit rate/reach capabilities than full DSL (target is 1 Mbit/s downstream) and this could vary markedly depending upon the quality of the customer's home wiring and the number and type of phones used. It is also highly likely that telephony performance will be affected. DSL-Lite is heavily supported by the PC and modem industry and may well be the push that the DSL market is needing. However, there are a number of potentially serious technical problems which must be overcome before it becomes a truly widespread viable technology. Optionally, the DSL modem could be inserted in the customer's PC; however, this may have regulatory impacts within the UK as it would require a 'copper-wire' interface to be specified.

- Traffic management

DSLAMs can potentially support hundreds of customers multiplexed on to one network interface and therefore improved traffic management capabilities are required to support different qualities of service and ensure fair sharing of bandwidth.

- Call control

Current moves are away from permanently set up connections towards connections set up on customer demand to better utilise core network bandwidth. The exact role the DSLAM plays in the call set-up and its ability to support signalling is a hotly debated topic and will take some time to resolve.

- Operational functionality

Operational functionality can be vastly improved over the capabilities offered in today's equipment through highly automated operations. These include streamlined configuration, comprehensive fault diagnostics and performance monitoring.

- Equipment dimensioning

Higher levels of integration will lead to reduced size and power dissipation of the equipment and ultimately unit costs.

The pace of development is rapid and it is expected that products will become available with some of these features in the very near future.

8. Conclusions

This paper presents some of the reasons why incumbent telcos have been seriously evaluating the use of ADSL to deliver broadband services. The basic technology has been proven to work through many trials throughout the world but just demonstrating the technology is not enough — it is necessary to develop a viable business case. As long as the equipment functionality remains basic and the market remains uncertain, the manual processes and relatively high unit costs are acceptable for niche applications. However, if the market develops and mass roll-out across the UK is required, then a much more sophisticated, scalable and cost-reduced solution is required than is available today. If DSL is going to rise to meet this challenge, a significant amount of development and investment will be required over the next few years.

The rate at which this will happen is debatable. It is the age old question — will it be technology push or market pull which will drive DSL roll-out?

Acknowledgements

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Alister Dickie joined BT in 1995 and has worked in the area of broadband communications for 12 years. This has involved leading major RACE and ESPRIT projects concerned with access and broadband switching and subsequent development of equipment based on the results of this work. His current role is as access broadband design authority where he is involved in several major projects looking at evolving BT's access network. This covers both near and long-term developments and the use of ATM, SDH, xDSL and PON technologies to provide a suitable evolution

path from current service demands to those expected to appear in the next 5—10 years. He has an honours degree in Electrical Engineering obtained from Salford University.



Jim Mackenzie joined BT in 1986 after gaining an honours degree in Applied Physics from Strathclyde University, Glasgow. Initially he worked on high-speed optical transmission system development which led to his involvement in the development of BT's SDH architecture and equipment specifications. For several years, he was very active in standards forums (ETSI, ITU), making many significant contributions to SDH architecture, protection and synchronisation work groups. For two years he led the ETSI working group on network synchronisation. In 1995, he moved to the

transport design and performance unit, to develop network architectures for future broadband services using ATM, SDH, PON and xDSL technologies. He now leads a team, developing BT's end-to-end transport network design for broadband services, including the designs for several internal BT trials.